

Dikshant Neupane

🎓 BSc. CSIT, 3rd Semester | 📍 Kathmandu, Nepal

✉️ dikshantneupane690@gmail.com • 📞 +977 9803391656 • 🌐 github.com/Dikshant-Neupane

1Professional Profile

BSc. CSIT student and self-driven AI practitioner building a portfolio that spans machine learning, scientific computing, algorithmic problem-solving, and full-stack development. Approaches problems with mathematical rigour — from implementing regression pipelines and chaos-theory simulators to designing community-impact platforms — and communicates findings with the clarity of someone equally at home writing code and writing prose. Driven by a belief that the most valuable data science work is both technically sound *and* visibly meaningful to the people it serves.

1Education

Bachelor of Science in Computer Science & Information Technology (BSc. CSIT)

2025 – Present

Samriddhi College, Kathmandu, Nepal

Currently: 3rd Semester

Core coursework: Data Structures & Algorithms, Object-Oriented Programming, Numerical Methods, Probability & Statistics, Database Management Systems, Software Engineering, Discrete Mathematics.

Independent study: Linear algebra for ML, advanced Python, Rust (in progress).

1Projects & Open-Source Work

House Price Prediction

[🔗 View Repository](#)

Machine Learning · Regression · Real Estate Analytics | Python · scikit-learn · pandas · Jupyter Notebook

- Designed and trained a supervised regression pipeline to predict residential property prices on a structured real-world dataset, achieving competitive error metrics via systematic model comparison.
- Conducted thorough **feature engineering** — encoding categorical location variables, handling skew, and creating interaction terms — to extract maximum signal from raw tabular data.
- Evaluated **multiple regression algorithms** (linear regression, decision tree, gradient boosting) using stratified cross-validation; selected the best model by MAE and RMSE against a held-out test set.
- Visualised prediction residuals and feature importances using Matplotlib and Seaborn to provide model interpretability alongside raw performance numbers.

Chaos Laboratory

[🔗 View Repository](#)

Scientific Computing · Dynamical Systems · Visualisation | Python · SciPy (ODE solvers) · Matplotlib

- Built an experimental Python library for exploring **chaos theory and nonlinear dynamics** through numerical simulation, implementing multiple classical systems (Lorenz, logistic map, Rössler).
- Integrated **ODE solvers** (Runge-Kutta via SciPy) to numerically integrate continuous dynamical systems with configurable precision and time horizons.
- Implemented **sensitivity analysis** routines to demonstrate divergence under infinitesimally small perturbations — visualising the butterfly effect through phase-space trajectory plots.
- Bridges pure mathematics and computational practice; the numerical methods used (Jacobian estimation, Monte Carlo-style perturbation) directly underpin probabilistic and simulation-based AI techniques.

Sahayog Fund — Community Crowdfunding Platform

[🔗 View Repository](#)

Full-Stack Development · Social Impact · Product Engineering | TypeScript · Full-Stack Web Application

- Designed and developed a **full-stack crowdfunding platform** aimed at community-driven, social-impact fundraising within Nepal, handling the complete product lifecycle from concept to deployment.
- Built core application features end-to-end: **user authentication**, campaign creation and management, fund-tracking

dashboards, and responsive UI — demonstrating product thinking alongside engineering execution.

- Implemented type-safe backend logic and API layer in **TypeScript**, applying software-engineering best practices (modular architecture, error handling, input validation).
- Reflects a deliberate interest in *technology for social good* — applying engineering skills to real community problems rather than purely academic exercises.

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Arbitrage Detection Tool

Algorithmic Problem-Solving · Quantitative Analysis · Graph Theory | Python · Graph Algorithms

• *Financial Data*

- Built a Python tool to detect **arbitrage opportunities** across multi-market price data by modelling price relationships as a weighted directed graph and identifying profitable cycles.
- Applied **Bellman-Ford negative cycle detection** (or equivalent logarithmic transformation) to systematically find arbitrage paths, translating a quantitative finance problem into a graph-theoretic algorithmic solution.
- Demonstrates the ability to map a real-world domain problem onto a mathematical structure and implement an efficient algorithmic solution with practical, verifiable outputs.

[View Repository](#)

Nepal Election 2082 — Multi-Factor Political Data Analysis

Political Data Science · Machine Learning · Statistical Analysis · Visualisation | Python · pandas

• *scikit-learn · SciPy · Matplotlib / Seaborn*

- Designed and analysed a **9-table relational dataset** spanning 55 constituencies, 26 parties, 215 candidates, and 3,000+ voters — covering candidate profiles, voter demographics, campaign rallies, social media trends, media coverage, and political scandals.
- Conducted **20 structured analysis sections** end-to-end: data loading, schema validation, Bikram Sambat date handling, referential integrity checks, and categorical encoding across all tables before any modelling.
- Applied **inferential statistics** (Mann-Whitney U, Pearson and Spearman correlation, linear and logistic regression) to quantify the measurable effects of social media sentiment, media coverage, campaign spending, and scandal exposure on vote share and win probability.
- Built a **Random Forest model** combining all multi-source features to predict each candidate's vote percentage; extracted feature importances to identify which signals — spending, sentiment, or scandal — most strongly drive electoral outcomes.
- Produced a full visualisation suite: parliament composition donut chart, winner vote-percentage by constituency, feature importance bar chart, and actual-vs-predicted scatter plot for model evaluation.

1Experience

2024 – Present

Communication & Marketing Lead (Volunteer)

Hult Prize at Samriddhi College

- Leads all written communication for the chapter: scripts, social media captions, event copy, and press content — developing a concise, audience-aware voice for a student social-enterprise audience.
- **Integrates AI tools** (LLMs, voiceover generators) into the content production workflow, actively optimising prompts to improve output quality and turnaround speed — hands-on applied prompt engineering.
- Coordinates across design, logistics, and event teams under tight deadlines, developing project-management and cross-functional collaboration skills.
- Translates technical and complex concepts into accessible, compelling narratives — a transferable skill directly applicable to AI research communication and public outreach.

1Technical Skills

Languages	Python (primary), TypeScript / JavaScript, C, Rust (learning)
Data & ML	pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Jupyter Notebook
Scientific Computing	SciPy (ODE integration), nonlinear dynamics, numerical methods, graph algorithms
AI & LLMs	Supervised regression, EDA, feature engineering, cross-validation, prompt engineering, applied LLM workflows
Dev Tools	Git, GitHub, VS Code, Linux / Bash, collaborative workflows

1Continuous Learning

- **Mathematics for ML:** Actively studying linear algebra, calculus, and probability theory as rigorous foundations for machine learning and AI.
- **Algorithms in C:** Implementing classic data structures and algorithms from scratch in C to build low-level intuition and strengthen algorithmic problem-solving fundamentals.
- **Rust:** Learning systems-level programming for performance-critical applications, with a focus on memory safety and concurrency.